AI Powered CCTV Monitoring & Compliance Platform

Project Understanding, Current Capabilities, Requirements & Next Steps

1. Executive Summary

The project is focused on developing a centralized AI powered monitoring platform that uses CCTV camera streams to detect, analyze, record, and report activities and compliance conditions within operational environments.

The existing AI and backend team has already developed a number of AI capabilities and supporting backend services. These include human detection, compliance detection, object detection, multi object tracking, facial recognition, event generation, database integration, alerts, and other AI based use cases.

The next stage is to transform these existing capabilities into a complete product through a frontend dashboard and a clearly defined API based integration.

The overall architecture can be understood as: CCTV Cameras → AI and Computer Vision Services → Backend APIs → Database and Events → Web Dashboard → Users, Reports and Alerts.

2. Project Objective

The primary objective is to create a platform through which users can monitor operational environments using existing CCTV infrastructure and AI models.

3. Existing AI & Backend Capabilities

A significant portion of the technical foundation already exists. The backend has been designed around AI services and microservices rather than a single monolithic application.

3.1 Human Detection

The system can detect humans from CCTV streams. Human detection can support occupancy monitoring, people counting, activity monitoring, surveillance, compliance monitoring, and further processing through facial recognition.

3.2 Compliance Monitoring

The AI system can detect various compliance related conditions including gloves, masks, hair nets, people, cars, fire and smoke, garbage, preparation areas, and other configurable objects or conditions.

Different detection classes can potentially be enabled for different cameras or areas.

3.3 Object Detection & Tracking

The backend supports object detection and multi object tracking. This provides the foundation for continuously analyzing camera streams and maintaining continuity around detected objects and people.

3.4 Facial Recognition

A facial recognition and surveillance capability has already been developed internally. The general process is Human Detection → Facial Feature Detection → Identity Matching → Known Person or Unknown Person.

The system can use an image database containing known employees and associate detected individuals with identities. It can recognize people from different angles, including side angles, and can connect identities with backend or CRM information.

4. Facial Recognition Requirements & Limitation

Facial recognition is technically available, but its effectiveness depends heavily on the camera stream. The camera needs sufficient resolution, clear facial detail, appropriate positioning, suitable distance, acceptable lighting, and a sufficiently visible face.

A person may therefore be successfully detected as a human while remaining unidentified. A suitable camera stream from the relevant Global IT environment was still required for deployment and testing of this capability.

Consent was also raised as an important consideration when using facial recognition in a customer environment.

5. Pizza Quality Monitoring

Pizza quality monitoring is another AI use case being developed. The AI team has been collecting and annotating images and training an initial model.

Potential pizza related detections discussed include pizza characteristics, toppings, beef versus chicken, burnt condition, and other quality attributes.

This area is less mature than some of the existing compliance capabilities and is expected to continue improving as additional real world training data is collected and annotated.

6. Importance of Real World Camera Data

One of the biggest technical challenges identified across the meetings is camera and data quality. The AI models depend heavily on the actual camera streams that will be used in production.

Issues discussed include low image resolution, 5 to 6 MP cameras or lower, blurry footage, distance between camera and subject, camera positioning, lighting conditions, insufficient facial detail, and lack of suitable views of preparation areas.

Camera configuration is therefore not simply an infrastructure issue. It directly affects AI performance and should be treated as a project dependency.

7. Data Annotation & Model Improvement

The AI team is actively building and improving datasets through image collection, annotation, dataset preparation, model training, testing, and improvement.

Label Studio has been used for manual annotation. This requires significant effort because the training data needs to represent the actual production environment.

The model is expected to continue improving as additional data becomes available. Model improvements should not require changes to the frontend. The backend and API interface should remain stable while the AI model becomes more accurate.

8. Backend Architecture

The backend follows a microservices oriented architecture. Different capabilities can run as separate services in Docker containers and communicate through the internal Docker network. The exact deployment architecture can

be finalized later.

9. API Integration

The frontend is expected to communicate with the backend through APIs. The backend team has indicated that endpoints should remain stable while AI models are improved.

The frontend team therefore needs to define what information it requires from the backend and establish the required API contract.

10. Frontend / Web Dashboard

The existing frontend should be treated as a functional proof of concept rather than the final production user experience. The next phase is to design and develop a proper user interface.

The frontend requirements still need to be formally defined.

10.1 Dashboard

High level operational statistics may include total cameras, active cameras, compliance status, active violations, recent events, alerts, people detected, unknown persons, and other AI statistics.

10.2 Camera Monitoring

Users should be able to navigate between cameras and view relevant monitoring information such as camera name, location, status, live or recorded stream, active detection types, current events, and compliance status.

10.3 Violations / Events

A dedicated area should allow users to review AI generated events including date and time, camera, location, event type, detection, person where available, status, severity if defined, and evidence where available.

10.4 Alerts

Users should be able to see important events requiring attention. The exact notification mechanism and business rules remain to be finalized.

10.5 Reports & Statistics

The system should provide meaningful statistics such as compliance trends, violations by camera or location, event frequency, people and occupancy statistics, detection statistics, and historical trends. The exact statistics need stakeholder approval.

11. Camera Configuration

The system should support configurable detection behavior per camera or area. Different areas may require different rules. The frontend should provide an appropriate configuration interface while the backend exposes the corresponding configuration APIs.

12. Events & Violation Management

The backend can generate events when AI detects a particular condition. An important business rule remains to be finalized: how repeated detections should be handled.

If a person remains without gloves for several minutes, the system may detect the same condition repeatedly. Without a buffer or debounce rule, this could create excessive duplicate alerts.

The product requirements should define what constitutes a new violation, how long a condition must persist, when another alert should be generated, when an existing violation is resolved, and whether repeated detections should be grouped into one incident.

13. Search & Filtering

The system should provide user friendly searching and filtering. Users should not need to interact with SQL directly.

Potential filters include date, time, camera, location, event type, compliance type, person, and status.

14. Audit Trail

An audit trail is required for important system actions. It should record who changed something, what was changed, when it was changed, the previous value, and the new value.

15. Users, Roles & Permissions

Role based access control was discussed as part of the platform. Different users may require different levels of access. The actual roles and permissions need to be finalized as part of the requirements.

16. Human Occupancy & Engagement

Another existing capability involves monitoring people in a call center environment. The system can detect people, track people, recognize known individuals where camera quality allows it, associate identified individuals with backend or CRM information, and support occupancy monitoring.

The concept of measuring engagement was also demonstrated. The exact definition of engagement and the statistics that should appear in the product need to be clarified before this becomes a formal product requirement.

17. Deployment & Environment

The deployment model has not been finalized. The existing team currently has a staging environment where multiple projects are running. The GPU environment is also used internally and cannot simply be exposed externally.

Two broad deployment possibilities were discussed: the existing team provides an appropriate staging environment, or the backend team packages and deploys the required services into the other team's staging environment.

18. Commercial & Ownership Considerations

Technical integration cannot be finalized independently of commercial discussions. Approval from the relevant client delivery, marketing, and management stakeholders may be required before sharing code, providing environment access, deploying services, or moving code to another staging environment.

19. Current Project Status

The project has a substantial AI and backend foundation, while the product frontend, exact requirements, deployment model, camera readiness, and some business rules remain to be finalized.

20. Key Project Dependencies

- Suitable camera streams are required for reliable AI performance, particularly facial recognition and detailed visual classification.
- Camera resolution, lighting, distance, positioning, and visibility need to be evaluated against AI requirements.
- The frontend team needs to formally define what users need to see and do.
- Once frontend requirements are defined, the required backend APIs and response structures should be documented.
- The technical deployment model must be agreed.
- Code and environment sharing arrangements require the appropriate internal approvals.
- Rules for alerts, repeated violations, incident grouping, and other operational behavior need stakeholder decisions.

21. Recommended Requirements Definition Process

Before extensive frontend development begins, both teams should align on a structured requirements document. For every feature, define the user, purpose, UI, user action, required data, API, business rules, and exception handling.

22. Suggested Initial Product Structure

```
AI Monitoring Platform
|
+-- Dashboard
|
+-- Cameras
|   +-- Camera List
|   +-- Camera Details
|   +-- Camera Configuration
|
+-- Live Monitoring
|
+-- Compliance
|   +-- Current Status
|   +-- Violations
|   +-- History
|
+-- Alerts
|
+-- People
|   +-- Detected People
|   +-- Known People
|   +-- Unknown People
|
+-- Pizza Quality
|
+-- Reports
|
+-- Users & Roles
|
+-- Audit Logs
|
+-- Settings
```

23. Overall Understanding

The project should not be viewed as simply building a frontend. There are three connected layers.

- Physical Infrastructure: cameras, camera placement, resolution, lighting, and streams.

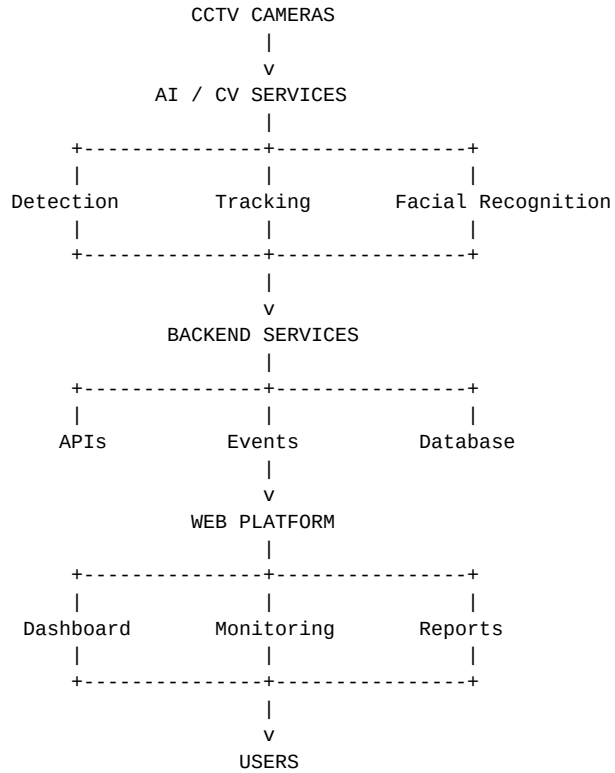
- AI and Backend: detection, tracking, facial recognition, compliance, pizza quality, events, database, APIs, and alerts.
- Product: dashboard, camera monitoring, violations, alerts, reports, configuration, users, and audit logs.

All three layers need to work together. A highly capable AI model cannot produce reliable results if the camera cannot provide usable images. Likewise, a strong backend does not automatically become a usable product without clear requirements and an effective frontend.

24. Immediate Next Steps

- Finalize product requirements covering dashboard, cameras, compliance, violations, alerts, people and facial recognition, pizza quality, reports, user roles, audit logs, and search and filtering.
- Define API requirements for each frontend feature, including endpoints, parameters, response structures, authentication, error handling, and real time requirements.
- Resolve infrastructure questions covering camera specifications, placement, streams, facial recognition stream, lighting, and whether cameras need replacement or repositioning.
- Resolve deployment and commercial questions covering service location, GPU processing, environment ownership, code and service transfer, and required approvals.
- Build the production UI once requirements and APIs are sufficiently clear.

25. Final Project Picture



Conclusion

The project already has a substantial AI and backend foundation. The main transition now is from AI proof of concept and backend capabilities toward a complete, usable product. The critical task for the next phase is to establish a clear bridge between the existing backend and the product frontend: Business Requirements → UI and UX → API Requirements → Backend Integration → AI Output → User facing Product.